

Mental Health Prediction using Machine Learning Models and Large Language Model

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Abstract - Mental health is integral to overall well-being, impacting human ability to deal with challenges in life. Machine learning and AI hold promise in predicting mental illnesses by analysing behavioural patterns, aiding in early detection and intervention. This proactive approach can mitigate symptom escalation, improving mental health outcomes. This research study introduces a novel predictive framework integrating ensemble learning techniques and large language models (LLMs). Initially, ensemble learning, including AdaBoost, voting, and bagging, constructs a robust model with Random Forest emerging as optimal. Subsequently, a Large Language Model (LLM) enhances the pipeline. User input triggers mental health prediction by Random Forest, forwarded to a Google Gemini model via an API key, generating personalised insights, marking a significant advancement in mental health prediction.

Keywords - Mental Health Prediction, Machine Learning algorithms, Artificial Intelligence(AI), Large Language Model (LLM), Early Detection In Behavioural Patterns.

I. INTRODUCTION

Mental health is a crucial aspect of overall well-being, requiring the same level of attention as physical health. From mild stress and anxiety to severe conditions like depression and bipolar disorder, mental health issues vary widely and can significantly impact one's life. In India, mental health remains a highly focused issue, adding to the challenges of addressing these issues effectively.

A mental health prediction platform is a valuable resource for individuals to stay informed about their mental well-being. By leveraging machine learning algorithms to analyse data such as sleep patterns, social media activity, and exercise routines, this platform can provide personalised insights and tools to support mental health care. The benefits are substantial, enabling proactive steps to improve mental well-being, identifying potential issues early, and facilitating access to professional treatment when needed.

Early detection and management are crucial for preventing mental health challenges from escalating into serious illnesses. Real-time applications of AI technologies can help identify individuals exhibiting unusual behaviour patterns, which may indicate underlying mental health issues. Utilising AI for mental health monitoring can proactively assist those in need, complementing traditional healthcare approaches and offering timely interventions to mitigate risks.

Overall, mental health significantly influences emotional, psychological, and social aspects of life, making it essential to address and manage effectively at all stages of life. By integrating AI technologies into mental health care, surveillance is enhanced, early detection, and support systems, ultimately improving outcomes for individuals facing mental health challenges.

II. LITERATURE REVIEW

Many studies applied machine learning methods to predict mental health disorders among employees using data from the OSMI mental health survey 2019. Algorithms like Logistic Regression, KNN, Random Forest, SVM, and Naïve Bayes are used. The study identifies logistic regression and decision tree classifiers as the most accurate in predicting mental health disorders, particularly influenced by family and personal mental health history in both technical and non-technical companies [4].

Researchers have explored data mining's potential in predicting mental wellness, using the OSMI survey data, Decision Tree and Random Forest algorithms. The study develops a website for users to input information and receive mental health predictions, achieving 82% accuracy with Decision Tree. Authors highlight mental health's importance and the system's potential to improve social and professional lives, suggesting further research to enhance specific mental illness predictions [6].

Multimodal educational data fusion is studied for detecting students' mental health issues using Deepwalk, MANE, SVM, RF, and XGBoost algorithms. Stress is monitored via wearable devices capturing physiological signals, with Python 3.6, Pandas, Scikit-learn, Origin 2018, and Graph used for analysis. Proposed approaches like CASTLE framework and MOON algorithm show promise in identifying mental health problems among students[8].

Some researchers have presented a system design for mental health identification using the "Mental Health Tracker" Android app and machine learning techniques like Natural Processing Language, Machine Learning, and MySQL for sentiment analysis. Psychiatrists review generated reports and administer necessary therapy based on findings, offering features like blogs, motivation, expert advice, exercises, professionals[12].

Artificial intelligence (AI) application in mental health analysis is explored using decision tree, logistic regression, and random forest models. The study emphasises comprehensive analytical processes for accurate prediction of mental health outcomes, urging further research for effective AI integration into mental health analysis [1].

An introduction of a machine learning approach for continuously monitoring patients' mental health crisis risk using electronic health records (EHRs) and algorithms like Logistic Regression, KNN, RF, and SVM. The model shows good crisis prediction performance in a clinical setting, enhancing outcomes and resource allocation in mental healthcare [16].

A framework using data science to address public mental health challenges, incorporating data mining techniques and decision tree algorithms. It stresses healthcare professionals' performance evaluation engagement for and successful implementation, emphasising data cleansing, feature selection, predictive modelling, and visual data exploration steps [17].

Machine learning techniques for forecasting mental health conditions, focusing on schizophrenia, anxiety, and depression. The study identifies prevailing algorithms, evaluates limitations, and outlines future research and intervention in ML-driven mental health prediction[14].

Some work explores machine learning techniques in identifying mental health issues, employing logistic regression, decision trees, random forest, support vector machines, and neural networks. The study evaluates classifier performance, with stacking showing high accuracy among classifiers [9].

The prediction of mental health issues among students using machine learning algorithms like Decision Tree, Neural Network, Support Vector Machine, Naïve Bayes, and Logistic Regression. The study balances interpretability and performance, with models trained to predict stress, depression, and anxiety levels among students [18].

Machine learning techniques to detect mental health issues using personal surveys and algorithms like logistic regression, KNN, RF, and SVM. Stress detection through physiological markers like heart rate and galvanic skin response is emphasised, highlighting the importance of innovative algorithms and mental health specialist input [3].

A study is carried out to employ the DISVM algorithm to identify depression tendencies among college students using Weibo data. The study evaluates algorithm efficiency and data selection for accurate depression recognition [13].

Mental health issues are studied among university students using algorithms like KNN, SVM, and Decision Tree. The study reveals high mental health issue rates among business students, with future research focusing on prediction model performance improvement [15].

The research predicts depression using machine learning algorithms based on work and personal factors from a personal survey dataset. Data visualisation and correlation matrices enhance public health by enabling early depression detection and treatment [11].

Analyses stress detection methods using wearable sensors and machine learning, emphasising cost effective stress monitoring models and potential for early problem detection[7].

III. PROPOSED METHOD

After collecting and integrating a dataset comprising 41,000 entries on mental health, all project dependencies were successfully imported and installed. Subsequently, the various models are trained and the performance is analysed.

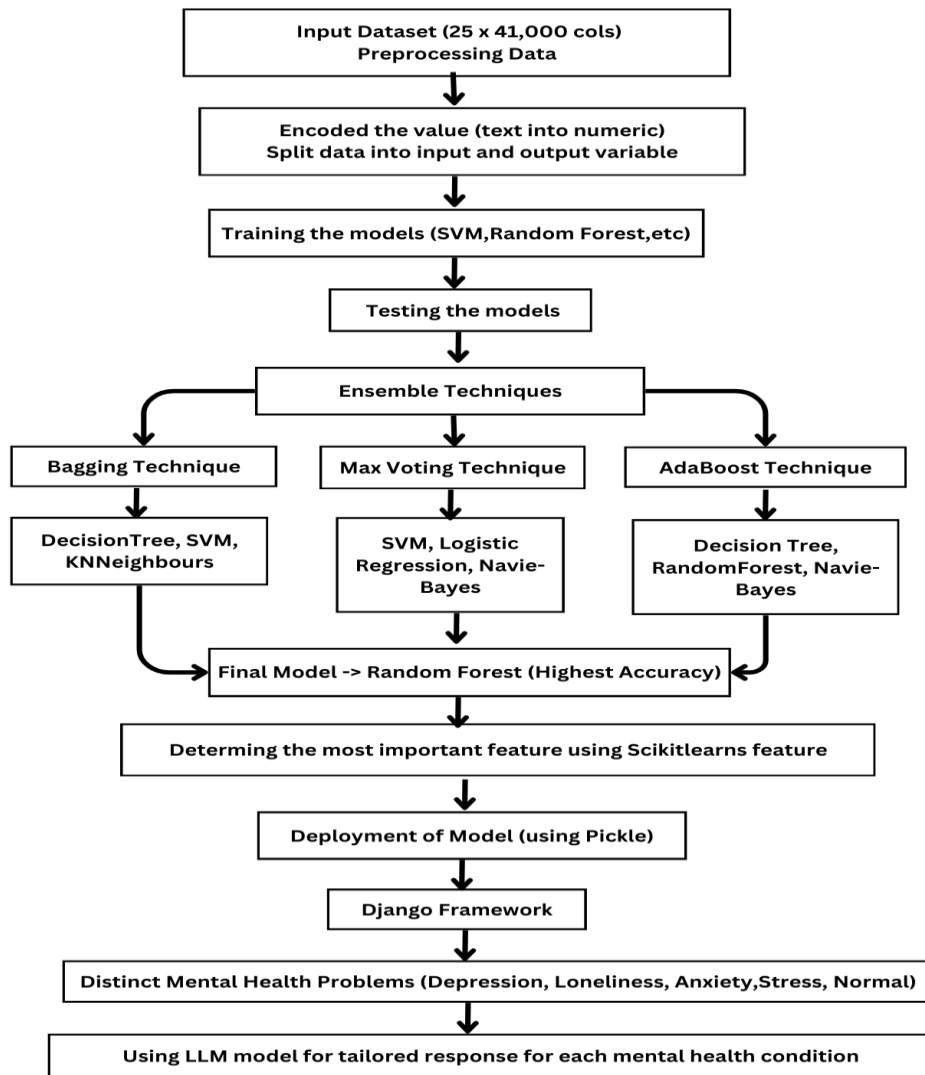


Figure 1. Flow diagram of Proposed System

Implementation involved testing several modules to evaluate accuracy. Specifically, the team employed a Voting Classifier, Random Forest Classifier, Support

Vector Mechanism (SVM), and Bagging Classifier to measure accuracy.

To evaluate the performance of machine learning models, particularly in classification tasks, the team constructed a confusion matrix. This matrix offers a comprehensive breakdown of true positives, true negatives, false positives, and false negatives, enabling a thorough assessment of the model's accuracy, precision, recall, and F1-score—all crucial metrics for understanding predictive performance.

Furthermore, ensemble techniques were employed to determine the optimal model. Through experimentation, the Random Forest model emerged as the most effective choice, demonstrating superior performance across various evaluation metrics.

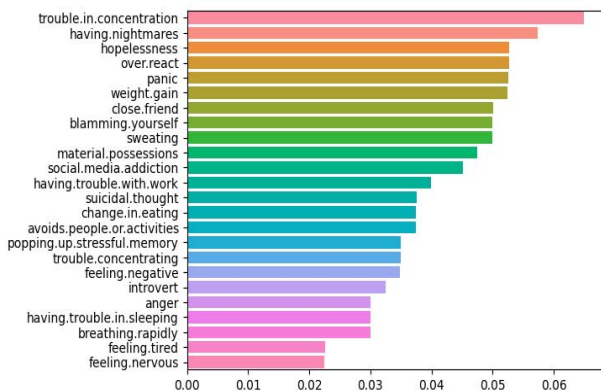


Figure 2. Feature Scores Image

IV. RESULT AND DISCUSSION

A. Existing System

- The Mind Therapy is a website designed to help people with mental health issues. It uses an interactive and user-friendly interface to offer various modules aimed at improving mental wellbeing. However, there are limitations to its functionality. Currently, it can only predict three mental health conditions: stress, anxiety, and depression. Additionally, the accuracy of its predictions might be compromised because some fields in its surveys are optional.
- Mental Health America offers a wealth of information through various quizzes that assess a wide range of mental health conditions. This versatility allows users to explore specific concerns. However, the website suffers from usability issues. The lack of clear navigation and visuals like pictures and videos can make it confusing for users seeking information or clarification about mental health conditions.

- Felicity stands out for its focus on user experience. The platform allows clients to leave reviews and access professional help through virtual consultations. Their counsellors offer a wide range of services to address various mental health concerns. However, the website could benefit from improved organisation. Currently, there's no dedicated section for specific features, and information on available counselling categories might be scattered, making it harder for users to find what they need.

B. Methodology

Feature engineering was meticulously conducted using the scikit-learn library, aimed at discerning crucial questions within the dataset to optimise the quiz framework. By leveraging advanced techniques within scikit-learn, significant features were identified, enhancing the efficacy and relevance of the quiz questions.

TABLE 1. COMPARATIVE RESULTS OF MODELS

Sr. No.	Model	Accu racy	Prec isio n	Reca ll	F1- Scor e	Sup port
1.	SVM	99%	0.98	0.99	0.99	818
2.	Logistic Regressi on	99%	0.99	0.98	0.98	821
3.	Random Forest	99%	0.99	0.99	0.99	826
4.	Bagging	99%	0.99	0.99	0.99	826
5.	Ada- Boostin g	98%	0.99	0.98	0.98	821

The aim of the research study is to create a predictive model for mental health assessment, a comprehensive ensemble approach was employed, integrating various techniques including AdaBoost, Voting, and Bagging. This ensemble methodology was designed to improve the accuracy of prediction. After experimentation and evaluation, the Random Forest model emerged as the most efficient predictive framework. Leveraging its inherent capacity for handling complex datasets and mitigating overfitting, the Random Forest model was seamlessly integrated into the predictive pipeline. Subsequently, to augment the predictive ability of the

model and enhance the user experience, an innovative augmentation was introduced through the utilisation of a Large Language Model (LLM). Upon user input of a query encompassing mental health-related concerns, the Random Forest model predicts the potential mental health issue. This predicted mental health issue is then forwarded to the Google Gemini model via an integrated Google API key. The Gemini model, leveraging advanced natural language processing techniques, dynamically generates descriptive

information based on the provided prompt. This augmentation not only enriches the predictive capabilities of the system but also facilitates personalised responses customized to the user's specific mental health concerns. The integration of the LLM module represents a novel approach in mental health assessment, effectively harnessing cutting-edge language processing capabilities to provide comprehensive and contextually relevant insights.

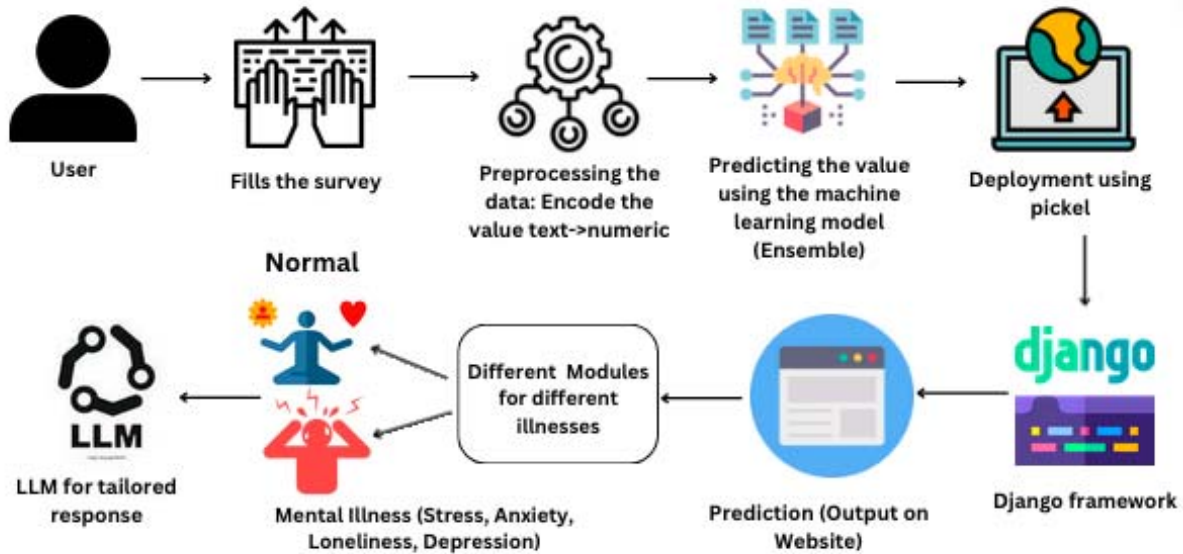


Figure 3. Architecture of the proposed model

C. Result

After the developed of machine learning model to predict mental health states using a dataset from a comprehensive survey with 41,000 entries, the model achieves remarkable accuracy. The trained and validated model demonstrates high precision in discerning various mental health conditions. Subsequently, the model was serialized into a pickle file for seamless integration and deployment.

Utilising the Django web framework, the team orchestrated the deployment of the model, embedding it within an intuitive front-end interface. This strategic integration of advanced technology and user-centric design principles positions the solution at the forefront of digital mental health care.

The model exhibits a profound capability to identify distinct mental states, including normalcy, stress, anxiety, loneliness, and depression. Leveraging a Language and Linguistics Model (LLM), the system creates responses to each detected mental health condition, providing users with informative and

empathetic feedback. The incorporation of an LLM enables the system to generate linguistically nuanced and contextually relevant responses, enhancing the overall user experience. By offering personalised insights and guidance, the solution empowers individuals to navigate their mental health journey with confidence and clarity.

V. CONCLUSION AND FUTURE SCOPE

The Mental Health Prediction platform merges technology, personalised assessments, and ethics to transform mental well-being. By advocating early intervention, reducing stigma, and fostering awareness, it aims to nurture empathy and understanding in society. Through a user-friendly interface and collaboration with professionals, the model empowers individuals to manage mental health effectively. Future enhancements may include integrating face recognition for more personalised services, reinforcing privacy, and promoting a culture that values mental well-being.

As part of future scope, integrating face recognition technology into the Mental Health Prediction platform could provide more personalised results and services. By analysing facial expressions and other biometric data, the system can offer tailored interventions and recommendations based on individual emotional states and patterns. This advanced feature not only enhances user experience but also improves the accuracy and efficiency of mental health assessments and interventions, further advancing the platform's mission of promoting early intervention and fostering a culture of mental well-being.

REFERENCES

- [1] C. Madhumitha, *Prediction Of Mental Health (Depression) Using Data Science And Machine Learning Techniques*, March 2022
- [2] Vaishnavi N Jadhav, Manasi S Bhamare, Manasvi U Vengurleka, Vidya S Kubde, *Mental Health Tracker*, www.Irjmets.Com/E-Issn:2582-5208, April 2022
- [3] Rohizah Abd Rahman, Khairuddin Omar,Shahrul Azman Mohd Noah1,Mohd Shahrul Nizam Mohd Danuri,Mohammed Ali Al-Garadi, *Application Of Machine Learning Methods In Mental Health Detection: A Systematic Review*, October 2020
- [4]Rohizah Abd Rahman, Khairuddin Omar, Shahrul Azman Mohd Noah1, (Member, Ieee),Mohd Shahrul Nizam Mohd Danuri (Member, Ieee), And Mohammed Ali Al-Garadi, *Predicting Mental Health Disorders Using Machine Learning For Employees In Technical And Non-Technical Companies*, October 6 2020
- [5] Dr.J.Arokia Renjit , Adlin Sajeesh M.J , Sangavai V.D , Sree Devi D.S, *Prediction Of Mental Health Using Machine Learning*, www.Jetir.Org (Issn-2349-5162), May 2022
- [6]Vidit Laijawala, Aadesh Aachaliya , Hardik Jatta, Vijaya Pinjarkar, *Classification Algorithms Based Mental Health Prediction Using Data Mining*, 10 July 2020
- [7]Shruti Gedam And Sanchita Paul, *A Review On Mental Stress Detection Using Wearable Sensors And Machine Learning Techniques*, June 16, 2021
- [8]Teng Guo, Wenhong Zhao(Member, Ieee), Mubarak Alrashoud, Amr Tolba(Senior Member, Ieee), Selena Firmin And Feng Xia (Senior Member, Ieee), *Multimodal Educational Data Fusion For Students' Mental Health Detection*, 30 June 2022
- [9] Konda Vaishnavi,U Nikhitha Kamath, B Ashwath Rao And Nv Subba Reddy, *Predicting Mental Health Illness Using Machine Learning Algorithms*, Year 2022
- [10]Manav Pradeep Jain, Soumya Prakash Sribash Dasmohapatra, Professor Stevina Correia, *Mental Health State Detection Using Opencv And Sentimental Analysis*, Year 2020
- [11] Shivangi Yadav; Tanishk Kaim; Shobhit Gupta; Ujjwal Bharti; Prakhar Priyadarshi, *Predicting Depression From Routine Survey Data Using Machine Learning*, 01 March 2021
- [12]Matthew Squires, Xiaohui Tao, Soman Elangovan, Raj Gururajan, Xujuan Zhou, U Rajendra Acharya And Yuefeng Li, *Deep Learning And Machine Learning In Psychiatry: A Survey Of Current Progress In Depression Detection, Diagnosis And Treatment*, 24 April 2023
- [13] Yan Ding, Xuemei Chen, Qiming Fu, Shan Zhong, *A Depression Recognition Method For College Students Using Deep Integrated Support Vector Algorithm*, 13 April 2020
- [14] Jetli Chung And Jason Teo, *Mental Health Prediction Using Machine Learning: Taxonomy, Applications, And Challenges*, 05 Jan 2022
- [15]Fazluddin Sahlan, Faris Hamidi, Muhammad Zulhafizal Misrat, Muhammad Haziq Adli, Sharyar Wani, Yonis Gulzar,*Prediction Of Mental Health Among University Students*, 2021
- [16] Roger Garriga, Javier Mas, Semhar Abraha, Jon Nolan, Oliver Harrison, George Tadros And Aleksandar Matic, *Machine Learning Model To Predict Mental Health Crises From Electronic Health Records*, June 2022
- [17]Charith Silva, Mahsa Saracee, Mo Saracee, *Data Science In Public Mental Health: A New Analytic Framework*, 27 January 2020
- [18]Sofiantia Mutalib, Nor Safika Mohd Shafiee, Shuzlina Abdul-Rahman, *Mental Health Prediction Models Using Machine Learning In Higher Education Institution*, 5 April 2021